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EMS502U
Thermodynamic Cycles

Analysis of Geothermal Binary Power Cycles Utilising both SRC and ORC

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1 Introduction

Steam Rankine Cycles (SRCs) have been widely used in power generation over history, ranging from coal powered to more modern approaches utilising nuclear heat generation. However, since the 1950s, when Organic Rankine Cycles (ORCs) were developed, a focus on implementing these as a form of heat recovery was established after determining their applicability in utilising low to medium temperature heat sources alongside high molecular mass organic fluids as the working fluid to generate electricity. This paper focuses on the analysis and comparison of SRCs against ORCs in geothermal binary powerplants, with an ultimate goal to conclude which is superior when working with low grade heat.

Geothermal energy is a renewable form of energy stored as heat within the Earth's interior, and both the natural steam and high temperature water form of it have been exploited for decades in many industrial processes (Barbier, 2002). However, unlike fossil fuels, geothermal sources provide a continuous supply of energy, reducing the dependency on non-renewable resources and promoting sustainability, additionally it is estimated that the energy stored at depth of 3 km from the Earth's surface is as high as $43,000,000 \times 10^{18}$ J (DiPippo, 2016). In 2000, the geothermal electrical installed capacity in the world was 7974 MWe (as seen in Figure 1) and the electrical energy generated was 49.3 billion kWh/year, representing 0.3 % of the world's total electrical energy which was 15,342 billion kWh. An increase of about 2GW was achieved in the five year term 2005–2010, with an estimate that the installed capacity of geothermal powerplants could make up 8.3% of total world electricity production by 2050 (Bertani, 2012). In recent times, ORC has raised great interest as it makes it possible to produce electricity from cooler geothermal sources, typically within the 80–150°C temperature range (Herath et al., 2020), often available from below 1000m (Cammarata, Cammarata and Petrone, 2014). Furthermore, the condenser temperature and pressure in geothermal binary powerplants can range from 10-40°C (Ghasemi et al., 2013), and standard atmospheric pressure of 1 bar, respectively. The standard first and second law efficiencies for geothermal binary plants were discovered to be approximately 5-15% (DiPippo, 2004) and 20-54% respectively (Franco and Villani, 2009), with the power output estimates around 2MW (Paweł Ziółkowski et al., 2021). The upcoming discussion will detail a critical analysis between SRCs and ORCs, as well as potential working fluid options for use within geothermal binary powerplants.

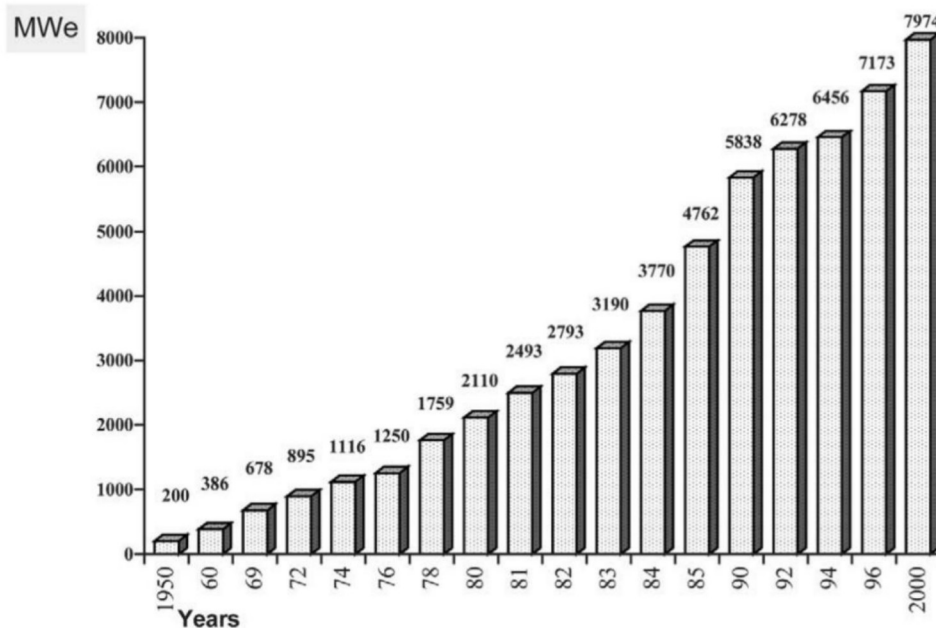


Figure 1: Illustration showing the increasing prevalence of geothermal power generation around the world during the period ranging from 1950-2000 (Barbier, 2002)

6 Appendix

6.1 Extended SRC calculations

Utilising Table 5, the h_f and h_g values for the isobar at 1.48 bar can be determined, and that of 10 bar can be obtained from the table.

Then, through following standard calculation procedures, h_1 , h_{1a} , h_3 , and h_4 can be calculated as follows.

$$h_1 = h_f(\text{at } 1.48 \text{ bar}) = 465.2 \text{ kJ/kg}$$

$$h_{1a} = h_1 + v_f(P_{1a} - P_1) = 465.2 + 1 \times 10^{-3}(10 - 1.48) \times 10^2 = 466.05 \text{ kJ/kg}$$

$$h_3 = h_g(\text{at } 10 \text{ bar}) = 2778 \text{ kJ/kg}$$

$$h_4 = h_f(1.48 \text{ bar}) + xh_{fg}(1.48 \text{ bar}) = 465.2 + \left(\frac{6.586 - 1.4294}{7.2276 - 1.4294} \right) \times 2227.2 = 2445.95 \text{ kJ/kg}$$

Therefore, using the methods displayed by Wang, Liu and Zhang (2014):

$$\dot{W}_{turb} = \dot{m}(h_4 - h_3) \times 0.85 = -6279.90 \text{ kW}$$

$$\dot{W}_{pump} = \frac{\dot{m}(h_{1a} - h_1)}{0.75} = 25.22 \text{ kW}$$

$$Q_{in} = \dot{m}(h_3 - h_{1a}) = 51440.89 \text{ kW}$$

Table 5: Properties table for saturated water and steam (Rogers and Mayhew, 1995)

Saturated Water and Steam

p [bar]	T_s [°C]	v_g [m ³ /kg]	u_f [kJ/kg]	u_g [kJ/kg]	h_f [kJ/kg]	h_{fg} [kJ/kg]	h_g [kJ/kg]	s_f [kJ/kg K]	s_{fg} [kJ/kg K]	s_g [kJ/kg K]
1.0	99.6	1.694	417	2506	417	2258	2675	1.303	6.056	7.359
1.1	102.3	1.549	429	2510	429	2251	2680	1.333	5.994	7.327
1.2	104.8	1.428	439	2512	439	2244	2683	1.361	5.937	7.298
1.3	107.1	1.325	449	2515	449	2238	2687	1.387	5.884	7.271
1.4	109.3	1.236	458	2517	458	2232	2690	1.411	5.835	7.246
1.5	111.4	1.159	467	2519	467	2226	2693	1.434	5.789	7.223
10	179.9	0.1944	762	2584	763	2015	2778	2.138	4.448	6.586

6.2 Extended ORC calculations

Through Table 6 it can be seen that the enthalpy values for the geothermal binary cycle utilising R600 have been precalculated by Basaran and Leyla Ozgener (2013), which were further verified by govinfo.gov (1982). Only paying attention to states 1, 2, 3, and 4, the operating conditions and corresponding enthalpy and entropy states. These calculations include the turbine and pump efficiencies of 0.85 and 0.75 respectively.

Table 6: calculated thermodynamic properties for R600 (Basaran and Leyla Ozgener, 2013)

State number	Working fluid	Phase	T (K)	P (MPa)	h (kJ kg ⁻¹)	s (kJ kg ⁻¹ K ⁻¹)	$\dot{m}$ (kg s ⁻¹)
0	Brine	Dead state	278.00	0.101	20.49	0.07398	–
0'	Cooling water	Dead state	278.00	0.101	20.49	0.07398	–
0''	R 600	Dead state	278.00	0.101	597.23	2.73010	–
1	R 600	Saturated liquid	283.00	0.148	223.04	1.08260	22.25
2	R 600	Liquid	283.53	1.000	224.97	1.08430	22.25
3	R 600	Superheated vapor	363.00	1.000	719.83	2.54170	22.25
4	R 600	Superheated vapor	311.95	0.148	649.94	2.58170	22.25
5	Brine	Liquid	369.00	0.500	401.98	1.25980	75.00
6	Brine	Liquid	334.00	0.500	255.14	0.84170	75.00
7	Cooling water	Liquid	280.00	0.500	29.29	0.10410	283.25
8	Cooling water	Liquid	288.00	0.500	62.83	0.22221	283.25

$$h_1 = 223.04 \text{ kJ/kg}$$

$$h_2 = 224.97 \text{ kJ/kg}$$

$$h_3 = 719.83 \text{ kJ/kg}$$

$$h_4 = 649.94 \text{ kJ/kg}$$

$$\dot{W}_{turb} = \dot{m}(h_4 - h_3) = -1555.05 \text{ kW}$$

$$\dot{W}_{pump} = \dot{m}(h_2 - h_1) = 42.94 \text{ kW}$$

$$\dot{Q}_{in} = \dot{m}(h_3 - h_2) = 11010.64 \text{ kW}$$